

Patterns in Paired Data



Grade 8 • Mathematics
NC.8.SP.1

Represent and describe bivariate associations.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

An invented study records practice sessions x and task scores y : $[(1, 18), (2, 15), (3, 14), (4, 11), (5, 10), (6, 7)]$. Plot every pair. Describe the association and explain why these data alone do not prove cause.

As x increases, y generally decreases. Points follow an approximate line. Other factors or selection of participants could explain the relationship.



Build a second model together

An invented study records practice sessions x and task scores y : $[(1, 24), (2, 25), (3, 28), (4, 29), (5, 32), (6, 33)]$. Plot every pair. Describe the association and explain why these data alone do not prove cause.

Discuss: What should the first step find?



Your turn • 1

An invented study records practice sessions x and task scores y :
[(1, 14), (2, 11), (3, 10), (4, 7), (5, 6), (6, 3)]. Plot every pair.
Describe the association and explain why these data alone do not prove cause.

Represent → solve → explain → check



Your turn • 2

An invented study records practice sessions x and task scores y :
[(1, 20), (2, 21), (3, 24), (4, 25), (5, 28), (6, 29)]. Plot every pair.
Describe the association and explain why these data alone do not prove cause.

Represent → solve → explain → check



Your turn • 3

An invented study records practice sessions x and task scores y :
[(1, 18), (2, 15), (3, 14), (4, 11), (5, 10), (6, 7)]. Plot every pair.
Describe the association and explain why these data alone do not prove cause.

Represent → solve → explain → check



Your turn • 4

An invented study records practice sessions x and task scores y :
[(1, 19), (2, 20), (3, 23), (4, 24), (5, 27), (6, 28)]. Plot every pair.
Describe the association and explain why these data alone do not prove cause.

Represent → solve → explain → check



Exit question • Explain independently

An invented study records practice sessions x and task scores y : $[(1, 16), (2, 13), (3, 12), (4, 9), (5, 8), (6, 5)]$. Plot every pair. Describe the association and explain why these data alone do not prove cause.

Include a check. What would you teach someone else?



Teacher review and next steps

1. Negative approximately linear association

Exit: Negative approximately linear association

2. Positive approximately linear association

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. Negative approximately linear association

4. Positive approximately linear association

